

sub-1-volt rails. The current industry standard, covering about 90 per cent, uses a three-stage conversion from 400 volts to 48 volts, then to 12 or 6 volts, and finally to the GPU core voltage. **Our current product, taping out in Q3 2026, focuses on this final stage.** For next-generation designs, we are exploring direct 800-volt-to-bus-voltage conversion and evaluating whether a one- or two-stage final conversion is more efficient.

A key challenge is the topology that enables such large conversion ratios (1000:1) in one step and identifying a control approach that balances efficiency with GPU load demand. C2i's architectural uniqueness lies in its software-defined VR approach.

For low-voltage stages, silicon is sufficient, but for high-voltage conversion, we partner with fabs that provide GaN solutions.

#### **At what level is the 10% efficiency gain measured, and where does it come from?**

The 10 per cent efficiency gain is measured at the system level and comes from the company's proprietary IP, with a reduced number of stages and optimisation across both control and power conversion in each stage. **The company has filed nine patents, with several more in progress, enabling extremely fast switching and highly optimised power conversion.** Efficiency losses in a system typically occur in three areas. First, during voltage conversion from the grid level to the core level, each step incurs losses. Second, intermediate bus routing on the motherboard, carrying hundreds to thousands of amps, introduces resistive losses. Third, GPU operation involves frequent transitions between high-performance and deep-sleep modes, where latency in power management can reduce performance (indirect efficiency).

The company's IP optimises across all three levels, resulting in a

net 10% gain in queries processed per watt.

#### **How would you sum up the advantages of your power management solutions?**

Our power management architecture offers several differentiators over incumbent solutions. **We are developing a software-defined voltage regulator that is highly scalable, operates from the grid to the core, and is processor-agnostic. It can support systems ranging from 100 watts to several kilowatts without compromising performance.** Key technical advantages include superior thermal efficiency, higher overall system efficiency, and compact density, reducing real estate requirements.

Additionally, the company's proprietary IP extends the lifespan of power solutions, effectively doubling the life of server power systems compared to existing components.

These innovations translate into measurable gains, including up to a 10% improvement in GPU performance (tokens processed per watt) and significantly longer server lifecycles.

#### **How do you approach co-design between power delivery and AI compute loads?**

Co-design involves two layers. For power delivery, we work closely with processor companies and OEMs because every server solution has unique requirements.

For hardware-software partitioning, our die currently runs only C2i Semi's own algorithms, both in hardware and software. We collaborate with customers to demonstrate the performance advantages of these algorithms.

If a feature meaningfully improves system performance, we enable it, allowing customers to benefit and potentially market it. Internal partitioning between hardware and software is driven by architectural choices and embedded

intelligence, refined through ongoing customer engagement.

#### **How did you validate the reliability of your design before committing to silicon?**

We validate reliability through close collaboration with fabs such as Tower and GlobalFoundries, reviewing designs and data with their process engineers on a fortnightly basis.

Beyond that, C2i's experience provides an edge. Our team has worked on similar products long enough to anticipate critical gaps that simulations cannot fully capture.

For areas that cannot be validated pre-silicon, we plan accordingly, ensuring budgets and design margins account for variations between silicon and simulation.

#### **What are the main risks between tape-out and first silicon validation?**

The biggest risk is schedule, especially since we are on a multi-project wafer. Any delay can push back silicon delivery, so we focus on completing all tasks on schedule while leaving time to verify changes.

Another challenge is the inherent uncertainty between simulation and silicon. Beyond the chip itself, factors such as packaging, package-to-motherboard interactions, and passive components introduce variations that are difficult to model accurately.

No one can guarantee 100 per cent performance at this stage, so we build in design and verification buffers to mitigate these risks and ensure the final product meets expectations.

#### **How does a startup like C2i Semiconductors protect its core IP while collaborating with customers or partners?**

We protect our core technology primarily through patents. We have already filed **nine** and have **several more** in the pipeline. For